

Element E:

Design Requirement	STEM Application	Metrics/Numbers	Expert reviewed comments
Educational Value	Interactive learning modules on aquaponics principles; allows for data collection for student analysis.	Measure the students average grades in science related topics before and after the aquaponics system was implemented.	Mr. Rickert - The students will benefit from the physical system, which helps students who learn that way.
Scale and Size	Optimized system size for educational impact and budget constraints.	System dimensions: 2m x 1m x 1.5m; designed to fit within a \$1000 budget.	Mr. Rickert - It is important that it will fit into the designated space given to us, and the dimensions provided will fit.
Safety Considerations	Use of non-toxic materials; secure and safe housing for electrical components.	All materials meet ASTM safety standards; electrical components rated for low voltage (12V).	Mr. Rickert- Safety is the most important thing in a school, and the use of non-toxic materials and low voltage electrical components will help with that.

Accessibility	Design accommodating students of various heights and abilities.	Adjustable viewing platforms; braille labels for plants for blind students	Mr. Rickert- There are many different students with a range of physical abilities, and they should all be able to benefit from the system. Adjustable viewing heights will accommodate that.
Sustainability	Use of recycled materials; efficient water and energy use.	At least 75% recycled materials; water recycling rate of 95%; solar-powered pumps.	Mr. Rickert - Recycled materials will benefit the environment, which is the whole purpose of aquaponics.
Monitoring and Control	Integration of sensors and control systems for educational monitoring.	pH, temperature, and nutrient sensors; control system with student interface for data analysis.	Mr. Rickert - Sensors and control systems will better help the students learn through the aquaponics system.

This table outlines the specific metrics and numbers that align with the STEM principles applied to the aquaponic system project. It’s designed to meet the highest standards of educational value, safety, accessibility, sustainability, and monitoring capabilities within the specified budget. By providing clear, quantifiable metrics, the project demonstrates a rigorous application of STEM practices.

Expert	Credentials
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Clint Rickert	- Masters degree in Technical Education and Agriculture from University of California, Pennsylvania - Bachelor's degree in Science Education from Widener University - Science and technical education teacher for 22+ years
Mr. Muscarella	- Bachelor's degree in Chemical Engineering from Drexel University - Master's degree in Science of Instruction - Resident Engineering teacher, teaching for 28+ years